

Line Follower Robot Project Portfolio

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Project Overview

This portfolio presents a team robotics project focused on the development, assembly, testing, and operation of a line-following robot. The robot uses infrared sensors and an ATmega64A microcontroller to detect and follow a predefined path autonomously.

My Contributions

- Soldered infrared sensor boards
- Soldered the main control board
- Participated in robot assembly
- Assisted with testing and troubleshooting
- Learned embedded systems and robotics concepts
- Collaborated as part of a robotics team

Technical Skills Demonstrated

Robotics, Electronics Assembly, Soldering, Embedded Systems, Microcontrollers, Infrared Sensors, PWM Motor Control, Troubleshooting, Teamwork, and Problem Solving.

Hardware Components

ATmega64A Microcontroller, Infrared Line Sensors, DC Motors, Motor Driver, Lithium Battery Pack, Resistors, Inductors, and Supporting Electronic Components.

Navigation Logic

The robot follows a black line on a light surface. Sensor data is processed by the microcontroller. Based on sensor readings, the robot moves forward, turns left, turns right, or stops when the path is lost. PWM signals are used to control motor speed and direction.

Challenges and Solutions

During development, hardware and soldering issues were encountered. Through testing, debugging, and teamwork, connection problems were identified and corrected. This process improved practical engineering and troubleshooting skills.

Learning Outcomes

This project provided hands-on experience with robotics, electronics, assembly, testing, and engineering workflows. It strengthened technical skills and increased understanding of embedded systems and autonomous robotics.

Future Improvements

Potential future enhancements include PID-based control, improved sensor calibration, faster navigation performance, and mechanical design optimization.

Portfolio Statement

This project represents an important step in my robotics journey. It demonstrates practical experience in electronics, teamwork, testing, and engineering problem solving through a real-world robotics project.